1. ECDLP lead pursuit audit

160-lead reconciled atlas · 2026-11-01



1.1. Decision

The 159-row ECDLP atlas at `crypto-autoresearcher@7072741f712c` was reviewed family by family against `crypto@b6ffdca53494`. The review also added one missing live split-graph/secant successor, so the reconciled atlas has **160 leads**. Exactly **21 narrowly scoped mechanisms** have mechanism-wide obstruction evidence under their stated assumptions. The other **139 leads remain pursued** through a next action or explicit revisit trigger.

This is a scope decision. Each stopped row is stopped only for the exact mechanism in its obstruction. A different presentation, locus, oracle, statistic, or stated escape remains live.

1.2. Queue invariant

Only `closed_obstructed` leaves the pursuit queue. `closed_known`, `closed_dominated`, and `exhausted_at_scale` remain queued through their revisit triggers. Timeouts, OOMs, missing infrastructure, unavailable sources, finite nulls, and finite measured cost disadvantages are not mechanism-wide disproof. A complete lower bound inside an explicitly named model obstructs only that model.

When a row combined a proved negative subcase with an untested escape, the audit preserved the obstruction as evidence and restored the row to `open_next_step` using its existing `revisit_when`. Twenty-six overbroad closures were corrected this way.

1.3. Family accounting

Family	Reviewed	Obstructed	Pursued
Binary-field IC	23	2	21
IC engines	20	2	18
Prime-field IC	27	3	24
Generic, transfer, structure, exotic	90	14	76
Total	**160**	**21**	**139**

1.4. Exact suppressed scopes

- Binary IC: `generic-table-oracle`; linear/subspace scope of `frobenius-orbit-factor-base`.
- IC engines: coordinate-blind scope of `generic-oracle-arity-cost-floor`; generic-fibre census of `semaev-cover-monodromy`.
- Prime IC: `generic-table-decomposition`; generic equality-filter scope of `representation-subset-sum`; `self-pairing-prime-order`.
- Generic: `generic-sqrt-lower-bound`; generic auxiliary-power acquisition in `cheon-aux-acquisition`.
- Transfer: `cm-canonical-lift-heights`; `one-point-self-pairing`; homomorphic scope of `padic-canonical-section`; class-invariant scope of `prime-field-isogeny-to-weak`; homomorphic scope of `second-torsion-direction`.

- Structure: generic scopes of `endo-scalar-action-generic`, `isogeny-class-generic-invariance`, and `isogeny-transport-preprocessing`; class-invariant weaknesses in `isogeny-walk-weak-neighbour`; scalar/generic scopes of `p192-class-cm-relations` and `secp256k1-cm-class-audit`.
- Exotic: the named GGM-simulable scope of `augmented-oracle-ggm`.

1.5. Load-bearing corrections

The binary SAT/enumeration measurements do not test symmetric Groebner systems at `ell=6..8`, high-arity closure, or the GG14 `m=4` presentation. The Semaev chain identity assumes rather than measures topology-independent closure degree. Finite volcano, endomorphism, factor-base, and residual-relation panels do not prove absence beyond their tested regimes. Proofs for homomorphic, bounded-degree, generically Galois, or full-Jacobian routes do not close explicitly non-homomorphic, growing-degree, non-Galois, or projected successors.

Load-bearing cross-checks are `crypto:research/factor_base_arity_review_20261024/README.md`, `crypto:research/prime_factor_base_degree_20261031/parallel_20261031/RESULTS.md`, `crypto:problems/ghs-higher-magic-descent-not-executable.md`, and every atlas row's obstruction, foreclosure, revisit, evidence, and related fields.

No performance result is introduced. This report records disposition and queue semantics only.

1.6. Review boundary

The Conductor command was unavailable and no `.conductor/WORKFLOW.md` was present. The global review stamp remains unchanged: the cryptanalysis base ref was unreachable and the configured `ecc-challenges` checkout was absent, so a whole-universe freshness claim would be unsupported. The 159 pre-existing rows, the added live successor, and the cited crypto snapshot are reconciled; the unavailable repositories must be reconciled before advancing the canonical stamp.